

# **The Impact of Blog Recommendations on Security**

## **Prices and Trading Volumes**

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Previous research and recent statements by the Securities and Exchange Commission indicate that e-mail spam and message board posts are often used to manipulate markets in a variation of the classical “pump-and-dump” scheme, leading to temporary market reactions followed by price reversals. In contrast, I hypothesize that financial blogs spread genuine information leading to permanent market adjustments.

I investigate stock recommendations on blogs and find that bloggers tend to write about liquid securities issued by large firms; I also find that bloggers offer short advice consistent with momentum strategies but long advice consistent with contrarian strategies.

To test the hypothesis that stock recommendations on blogs impact both prices and trading volumes of the touted securities, I collect recommendations from blogs and analyze returns and trading volumes on the days surrounding publication; I offer evidence to support my hypothesis and I find some evidence of market reaction being stronger for short recommendations. I hypothesize and test that the magnitude of the market reaction depends on the market capitalization of the touted firm, on the size of the blog’s audience, on the depth of the analysis of the blog post and on the perceived skill of the blog’s author. I find that the market appears to react more strongly to recommendations given by holders of a graduate degree in Finance or Economics; the other hypothesized factors do not have an impact on the magnitude of the reaction. Finally, I document the absence of price reversals in the twenty days following blog publication, giving support to the hypothesis that blogs offer genuine information.

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While recent and accurate estimates of blog<sup>1</sup> traffic are difficult to obtain, given the fragmentation of the medium, a study made public in August 2005 by marketing firms SixApart and Gawker Media claimed that 50 million Americans, or 30 percent of American internet users, visited a blog during the first three months of 2005. On August 24, 2006, Technorati, a website monitoring blog activity, reported that the blog by Chinese starlet Xu Jinglei had recorded an excess of 50 million page views, a new record. Technorati also lists in excess of 103 million blogs as active, accounting for over 250 million WebPages; the estimated rate of growth is of about 175,000 blogs per day, over 1.6 million posts per day, or over 18 blog posts per second<sup>2</sup>. All the above indicates that blogs are a major new-media phenomenon; now, they appear to be making their way in the financial advising sector. On Sept. 12, 2006, Yahoo! Finance announced it will enhance its finance site with a web log compiling blog posts from portfolio managers, hedge fund managers and other finance professionals. Google Finance, Yahoo's main competitor, already includes links to web logs.

Given the above, it is reasonable to question whether investors consider blogs a legitimate source of information; yet, an investigation into market reactions to blogs would not be complete without first attempting to describe core characteristics of financial blogs. Accordingly, prior to engaging the main topic of the present (describing market reactions to financial blogs), I pose and attempt to answer the following questions:

*Q1: How do bloggers select stocks to recommend?* I present descriptive statistics comparing recommended securities to the “average” security. In addition, I test whether bloggers write about stocks that have displayed abnormal returns or trading volumes in the recent past, attempting to determine whether bloggers follow momentum or contrarian trading strategies.

*Q2: Are stock recommendations given on blogs a valid source of investment advice?* I test whether recommended securities display abnormal returns after publication of the blog.

*Q3: Do blogs differ in terms of the quality of investment advice?* I investigate whether blogs differ in terms of quality of investment advice and whether differences in quality can be linked to specific characteristics of the post (depth of analysis), of the blogger (being a finance professional or a retired finance professional, a finance student, a journalist, possessions of a

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<sup>1</sup> “Blog” is a contraction of the term “Web Log”. Merriam Webster defines a “blog” as “a Web site that contains an online personal journal with reflections, comments, and often hyperlinks provided by the writer”.

<sup>2</sup> Data reported by Technorati Inc., available at: <http://technorati.com/about/>.

Certified Financial Analyst or equivalent designation, having earned a graduate degree in Finance or Economics) or of the blog (institutional affiliation and size of the audience).

I find that bloggers focus on liquid securities issued by large companies displaying abnormal returns and trading volumes in the recent past. In short recommendations, bloggers mostly follow a momentum strategy, while in long recommendations they appear contrarian. Blog recommendations are, on average, accurate<sup>3</sup>; accuracy of recommendations does differ by blog.

After presenting an analysis of the financial blog universe as outlined above, I return to the main question: do investors consider blogs a legitimate source of information? If so, I would expect a positive (negative) price reaction to buy (sell) recommendations as investors rush to buy (sell) the recommended securities. Similarly, I would expect abnormally high trading volumes following recommendations. This expectation is reinforced by previous studies that have found significant market reactions to financial advice posted on internet message boards (Wysocki, 1999, Tumarkin and Whitelaw, 2001, Dewally, 2003 and Antweiler and Frank, 2004) and delivered via unsolicited e-mails (Böhme and Holz, 2006 and Frieder and Zitztrain, 2006). Contrary to e-mail spammers and message-board posters, bloggers are not anonymous and have a clear incentive to spread genuine and reliable information, as the quality of the latter will reflect on their ability to attract a larger audience to their site. It is then reasonable to expect that investors would consider blogs a more reliable source of information and to expect similar, if not stronger, reactions; this leads to my first set of hypothesis:

*H1: Buy recommendations posted on blogs will lead to higher prices for the mentioned securities on the day of the posting and/or the following day; sell recommendations posted on blogs will lead to lower prices for the mentioned securities on the day of the posting and/or the following day.*

*H2: Both buy and sell recommendations posted on blogs will lead to abnormally high trading volumes for the mentioned securities on the day of the posting and/or the following day.*

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<sup>3</sup> I define and measure accuracy by comparing price movements to the recommended position; accordingly, a price increase (decrease) following a long (short) recommendation indicates an accurate recommendation. Under such a definition, accuracy is the cumulative effect of predictive ability and of market reaction to the blog itself. While I do not aim at distinguishing between accuracy and predictive ability of blogs, I offer evidence of the fact that blog posts are not only accurate, but contain predictive ability as well.

Numerous previous studies (for example, Womack 1996) have observed that markets react more strongly to sell recommendations than to buy recommendations; the explanation commonly given borrows from the field of psychology in hypothesizing an irrational loss aversion. Accordingly, I expect to observe a stronger market reaction to sell recommendations than to buy recommendations, both in terms of abnormal returns and trading volumes. Accordingly, I hypothesize:

*H3: Sell recommendations posted on blogs will lead to larger absolute-value abnormal returns and abnormal trading volumes on the day of the posting and/or the following day than buy recommendations.*

Secondarily, I intend to analyze how this hypothesized reaction varies according to characteristics of the touted stock, of the blog and of the blogger. Given that less information is available in regard to small companies, I hypothesize that recommendations of bloggers should have a stronger impact for firms with smaller market capitalization. In addition, I analyze whether the market reacts more to advice given by bloggers meeting certain qualifications, (in particular, being a finance professional or a retired finance professional, a student of Finance, a journalist, possessions of a Certified Financial Analyst or equivalent designation, having earned a graduate degree in Finance or Economics) as I expect investors to be more likely to act on recommendations originating from a source they consider trustworthy. Similarly, I expect a weaker reaction to recommendations offered by anonymous bloggers and I also expect a stronger reaction to recommendations given by bloggers who do not hold a position in the touted security, as that reduces the risk of conflicts of interest. In terms of blog characteristics, I expect a stronger reaction to recommendations originating from blogs published by institutions, rather than individuals, as the reputation of the institution offers additional certification of the blogger's skill and trustworthiness. I expect a stronger reaction to blogs with larger audiences, as the information content of the blog reaches more individuals. Finally, I test the relationship between market reaction and the level of analysis of the blog; I expect investors to be more likely to react to advice given on blogs containing a high level of analysis. This leads to my second set of hypothesis:

*H4: The impact on prices and trading volumes will be greater for stocks of firms with low market capitalizations.*

*H5: The impact on prices and trading volumes will be greater for stocks touted on blogs written by individuals possessing one or more of the following credentials: being a finance professional or a retired finance professional, a finance student, a journalist, possessions of a Certified Financial Analyst designation or other equivalent financial certification, having earned a graduate degree in Finance or Economics.*

*H6: The impact on prices and trading volumes will be weaker for stocks touted by anonymous bloggers.*

*H7: The impact on prices and trading volumes will be stronger for stocks touted by bloggers who do not hold a position in the touted security.*

*H8: The impact on prices and trading volumes will be greater for stocks touted on blogs with a larger audience.*

*H9: The impact on prices and trading volumes will be greater for stocks touted on institutional blogs.*

*H10: The impact on prices and trading volumes will be greater for stocks touted on blog posts offering more in-depth analysis.*

I further hypothesize that, while stock touting by e-mail spam is, oftentimes, an attempt at manipulating markets, blogs are not. The Securities and Exchange Commission is currently conducting “Operation Spamalot”, an attempt at identifying and punishing e-mail spammers. On March 8<sup>th</sup>, 2007, the SEC has suspended trading on 35 companies due to concerns regarding market manipulation through e-mail spam<sup>4</sup>. In addition, Böhme and Holz (2006) and Frieder and Zittrain (2006) find evidence that financial advice in e-mail spam is consistent with existing theories of market manipulation (Aggarwal and Wu, 2003). In particular, they find strong evidence of price reversion and a pattern of abnormal volumes and returns prior to the delivery of e-mail spam, both consistent with the hypothesized manipulative intent. That is, e-mail spammers spread false information in order to move the price of a security; eventually, the price reverts to its fundamental value. In the case of blogs, it is plausible that posters are spreading genuine information in order to build a solid reputation; after all, while e-mail spammers reach their

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<sup>4</sup> Wall Street Journal, March 9th, 2007, “SEC Suspends 35 Small Stocks In Crackdown on Spam Hying”

audience by mass-mailing, bloggers have to attract readers to their sites; accordingly, I expect bloggers to be much more concerned about the quality of the information they provide. Hence, I hypothesize a lack of mean reversion in prices following blog publication:

*H11: There will be no price reversion for the recommended stocks subsequent to the initial reaction.*

I find that markets do react to blog recommendations, but while analysis of price movements offers clear results, the volume-based evidence has to be qualified, as abnormal volumes following blog publication are at least partially due to momentum. Bloggers do appear, for the most part, to echo information already available in the media. Yet, as market reactions to select blogs suggests, some do spread genuine information and analysis, which does lead to a market adjustment. Concerns about the potential for market manipulation are not motivated, as initial reactions are not followed by reversions.

The present research adds to the literature discussing how the market reacts to stock recommendations. More specifically, it adds to the niche research focusing on market reaction to “new-media” recommendations. Prior research has focused on e-mail spam and message board posts; this is the first study relative to blogs. From a regulatory perspective, I offer evidence of the fact that blogs spread genuine information and are not just another instrument for “pump and dump” schemes; given the recent interest by the SEC in regulating electronic media, I believe this result to be of great interest.

The paper is structured as follows: *Section I* is a discussion of the relevant literature and of the contributions of the present research; *Section II* is a discussion of the applied methodology and data; *Section III* presents results of the analysis; *Section IV* offers an interpretation of those results while *Section V* concludes.

## **I. Literature Review**

Market reaction to stock recommendations has been the subject of numerous studies. A first series of papers has focused on the effect of recommendations published on The Wall Street Journal on security prices and trading volumes. Davies and Canes (1978), Liu and Smith (1990), Beneish (1991) and Bauman, Datta, and Iskandar-Datta (1995) examine the impact of the "Heard-on-the-Street" (HOTS) column on The Wall Street Journal and conclude that stock prices adjust to revisions in analysts' recommendations. However, Liu and Smith (1990) and Beneish (1991)

also observe smaller, but statistically significant, price changes on the days preceding the publication, associated with higher trading volume. This abnormal activity prior to the stock recommendation suggests the possibility that the observed reaction might be due to the selection criteria used by the column authors, rather than to readers acting on the column's advice. Ferreira and Smith (1999) find similar results for stocks in the "Small Stock Focus" column in 1993. Sarkar (2000) looks at the impact of articles in regional Wall Street Journal Publications on stock prices. He observes that articles in regional publications have a statistically significant impact on stock prices on the day of publication and that the abnormal returns are smaller for articles discussing multiple companies than for single company articles.

Other studies have focused on recommendations originating from other printed media. Benesh and Clark (1994) find a large and statistically significant market reaction on the date of publication of stock selections on Barron's. Mathur and Waheed (1995) look at stock price reaction to the "Inside Wall Street" (IWS) columns from 1981-1989 in Business Week magazine. Their results are consistent with the HOTS articles studies. Kerl and Walter (2007) look at explicit buy recommendations published by the German Personal Finance Magazine, and find significant abnormal returns in the five days around the publication date. Yet, they conclude that those are due in part to the effect of the publication and in part to momentum.

A second line of research focuses on the impact of brokerage houses' recommendations. Groth et al. (1979) document abnormal returns following recommendations by a single investment firm; they conclude that those are due to the selection criteria employed by the brokerage. They also find that the reaction of stock prices is symmetric with respect to buy and sell recommendations, and the impact of single-company recommendations is greater than the impact of multi-company recommendations. Stickel (1995) analyzes buy and sell recommendations in the period 1988-1991. He finds short-term price reaction to be a function of the strength of the recommendation, the magnitude of the change in recommendation, the reputation of the analyst, the size of the brokerage house and the size of the recommended firm. Womack (1996) focuses on a smaller sample and similarly finds that there is strong evidence that stock prices are influenced by analysts' recommendation changes. He further observes, in contrast to Liu and Smith (1990), that such effects are stronger for sell recommendations than for buy recommendations. Francis and Soffer (1997) examine the relative information content of stock recommendations and earnings forecast revisions. They find that recommendations and forecast revisions both affect stock prices and that price reactions to buy and sell recommendations are stronger in the presence of a consistent forecast revision. Finally, Tafler and Ryan (2002)

examine the impact of a series of informational events on stock prices; they conclude that the strongest market reactions are linked to changes in analyst recommendations.

Two recent studies focus on analyst recommendations in the Italian Stock Market; Belcredi, Bozzi and Rigamonti (2003) and Cervellati, Della Bina and. Pattitoni, (2006) both find significant market reactions in both prices and trading volumes to changes in analyst recommendations. In contrast, Jagadeesh and Kim (2003) look at stock market reactions to changes in analyst recommendations in all G7 countries; they find significant market reactions in all markets except for the Italian Stock Exchange.

More specifically to online content, a first series of papers has analyzed the link between message board activity, trading volumes and short-term returns. While most research agrees on the existence of a correlation, studies have disagreed on the direction of such relationship; that is, whether abnormal market activity drives posts or whether posts lead to abnormal market activity is still up for debate. A first study focusing on the correlation between message board activity, trading volumes and short-term returns, was conducted by Wysocki (1999). He finds that cumulative posting volume is highest for firms that display high short-seller activity, high market valuations, low institutional holdings, high trading volumes, extreme performance and high analyst following. Without much evidence, he concludes that abnormal market activity drives posting. Similarly, Das and Sisk (2003) analyze over 23 million message board posts to examine how opinions about stocks are formed and how they spread. They find that stocks which are highly discussed display lower return variance and higher mean returns. A first study investigating the direction of the link between message board activity, trading volumes and short-term returns was presented by Tumarkin and Whitelaw (2001), who study message posting on the RagingBull.com discussion forum. They find that changes in investor opinion are correlated with abnormal returns on days with abnormally high message activity and that those event days also coincide with abnormally high trading volumes. However, they conclude that message board activity does not predict industry-adjusted returns or abnormal trading volumes. Consistently, Das and Chen (2001), analyze message board postings in the last quarter of 2000 and find a strong link between market movements and stock recommendations; they document that returns drive discussion on message boards. Similarly, Dewally (2003) examines stock recommendations given on message boards; he finds that posters tend to follow momentum strategies: that is, they recommend stocks after those have experienced sharp price increases. He finds that the stock market does not appear to react to those recommendations. The opposite conclusion, that message board postings drive returns and volumes on the stock market, is discussed by Antweiler and Frank (2004), who study message postings on Yahoo! and Raging Bull message boards for the

firms in the Dow Jones Industrial Average and the Dow Jones Internet Commerce Index during the year 2000. They find significant predictive content between message postings and trading volumes, between message posting and volatility and between the degree of bullishness and trading volumes. Consistently, Clarkson, Daniel and Tutticci (2006) examine market reaction to takeover rumor postings and find abnormal returns and trading volumes on the day before and the day of the posting, concluding that markets do react to rumor spread over the internet. While not focusing on message boards, Barber and Odean (2004) give strength to the idea that it is discussion to drive market behavior, rather than the opposite; they find that individual investors are net buyers of “attention-grabbing stocks”, defined as stocks in the news, stocks experiencing high abnormal trading volumes and stocks with extreme returns.

Recently, two contemporaneous researches have looked into the relationship between e-mail spam and stock reaction. A first paper by Böhme and Holz (2006) investigates the effects of e-mail spam on stock prices and trading volumes. Böhme and Holz find significantly positive cumulative abnormal returns and significant increases in trading volumes during the event window. Frieder and Zittrain (2006) investigate the same matter using different data and methodology and find similar evidence of significant positive returns and volume increases for touted stocks. Both studies find short-term mean reversion following the touts.

Aggarwal and Wu (2003) provide theory and empirical evidence for cases where stock prices have been manipulated. I plan on comparing price and trading volume patterns surrounding blog recommendations to those described in that research, to determine whether financial advice given on blogs is consistent with attempted market manipulation.

The basic methodology for event studies is described by Campbell, Lo and Mackinlay (1997). In addition to test statistics proposed in the latter, I check robustness of my results by employing a test statistic proposed by Boehmer, Musumeci and Poulsen (1991), accounting for event-induced variances.

The present research will add to the literature discussing how the market reacts to stock recommendations. In particular, as discussed above, prior research has focused on e-mail spam and message board posts, but this is the first study relative to blogs. Given the fact that bloggers have a clear incentive to build a reputation and that their profiles are available to the public, I expect investors to place greater faith in blogs than in e-mail spam or message board postings; as a consequence, I expect blogs to have an even more significant market impact.

## II. Methodology

The basic framework methodology for short-term event studies is described by Campbell, Lo and Mackinlay (1997). I fit a three factor model (Fama and French, 1993) to calculate excess, or abnormal, returns. All dates discussed in the present are relative to the event date, defined as the date of blog publication; consistently, I mark the date of blog publication as “day 0”. All other days are numbered accordingly. Since some blogs are published after market close, the market reaction, if any, will take place on day +1; similarly, it is easy to foresee that some blog posts will not be read immediately at the time of posting, thus leading to a similarly delayed reaction. Accordingly, I report and discuss results relative to the two-day event window including day 0 and day +1<sup>5</sup>. The estimation period employed is of 95 days ending 20 days prior to the event date. Tests for significance of abnormal returns are described by Campbell, Lo and Mackinlay (1997).

In order to investigate the link between blog posts and abnormal trading volumes, I estimate abnormal trading volumes; I use share turnover as a measure of daily trading volume, defined as the ratio between the number of shares traded during the day over the number of shares outstanding. This measure of trading volume has long been used in the financial literature (see, for example, Lo and Wang, 2000) and, in particular, in volume-based event studies (see, for example, Richardson and Sefcik, 1986 or Stickel and Verrecchia, 1994). I estimate abnormal trading volumes as the difference between the daily share turnover and median daily share turnover (as, for example, in Bamber, 1986, or Bamber, 1987); median share turnover is calculated over the previously defined estimation period (95 days ending 20 days prior to the blog publication)<sup>6</sup>.

I isolate a portion of the dataset into a subset controlled for news; that is, recommendations in this “controlled” subset are not concurrent with news events or other blogs which would justify market adjustments. I interpret abnormal returns and trading volumes associated with this controlled subset as a measure of the market’s reaction to the blog post.

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<sup>5</sup> Since it is impossible to determine whether most of a daily return on the day of a blog recommendation occurred before or after the blog posting, I attempt a different relative dating procedure. I arbitrarily define the event date as the date of blog publication if the blog has been published prior to mid-day and as the following day otherwise. In case of unavailable publication time, I use the date of blog publication as the event day, as most blogs in the sample do appear to be published in the early morning hours. This alternative dating procedure does not significantly alter any of the results discussed here.

<sup>6</sup> Daily trading volume has also been defined as the daily unadjusted number of shares traded and abnormal volumes have then been computed as the difference between the daily number of shares traded and the median number of daily shares traded (computed over the 95 days ending 20 days prior to the blog publication). All reported results are insensitive to this alternative definition of volume. Also, please note that Bamber (1986) documents that daily abnormal volumes calculated for 349 firms listed on either the NYSE, NASDAQ or AMEX for the years 1977 to 1979 are not sensitive to the use of a market-adjusted share turnover rather than the median-adjusted share turnover.

In the overall sample, cumulative abnormal returns are my measure of accuracy of blog recommendations (positive abnormal returns following long recommendations indicate accurate predictions; similarly, negative abnormal returns following short recommendations indicate accurate predictions); please note, when investigating accuracy of recommendations and market reactions, I multiply returns on the short subset by -1 in order to obtain comparable results for the long and short subset; cumulative abnormal returns adjusted in such a manner are used in both the analysis of variance and in the cross-sectional regressions discussed below. I use analysis of variance (a one-way ANOVA) on cumulative abnormal returns to investigate whether blogs differ in terms of quality of advice. Similarly, in the controlled sample, I interpret standardized cumulative abnormal returns and abnormal trading volumes as a measure of market reaction to the blog post; accordingly, I employ a one-way ANOVA on both 2-day abnormal returns and 2-day abnormal trading volumes on a subset of the dataset which contains only recommendations with no concurrent news to determine whether market reaction to certain blogs is stronger than to others.

I also present a cross-sectional regression of standardized cumulative abnormal returns against a set of explanatory variables. I compute and regress standardized cumulative abnormal returns for the 2-day event window (day 0 and day +1) and over a 20-day event window (day 0 to day +19) to take into account both short- and medium-term accuracy<sup>7</sup>. The set of explanatory variables include a binary variable indicating whether the recommendation is long or short, on the number of words in the blog post (proxing for depth of analysis), a measure of blog influence (Technorati Score), a series of blogger-specific binary variables (indicating whether the blogger is a finance professional or a retired finance professional, a finance student, a journalist, whether the blogger is anonymous or whether the blogger possesses a Certified Financial Analyst designation or other equivalent financial certification, whether the blogger has earned a graduate or doctoral degree in Finance or Economics and whether the blogger is investing in the recommended security) and a binary variable indicating whether a blog is institutional, in order to determine whether those factors predict abnormal returns and abnormal trading volumes following blog recommendations. Cumulative abnormal returns for the 20 days preceding the blog publication

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<sup>7</sup> Here and in the rest of the paper, when calculating cumulative abnormal returns (or standardized cumulative abnormal returns) for a period of time exceeding one day, I simply add the abnormal returns (standardized abnormal returns) for the days of the interval. For example, the 2-day event window CAR will equal the sum of the day 0 and day +1 abnormal returns; the 5-day event CAR will equal the sum of the abnormal returns for day 0, day +1, day +2, day +3, day +4 and day +5. Please note the difference between the 5-day event (including days 0, +1, +2, +3 and +4) and the 5-day post event (including days +2, +3, +4, +5, +6).

are added as an explanatory variable in order to control for possible momentum or reversal effects.

Similarly, I also present a cross-sectional regression of 2-day standardized cumulative abnormal returns and abnormal trading volumes of the “controlled” subset. The predictive variables I investigate are the set described in the previously mentioned cross-sectional regression plus an additional variable proxying for depth of analysis (whether the blog contains multiple recommendations<sup>8</sup>), a variable equal to mean market capitalization of the firm issuing the recommended security and a binary variable identifying posts by anonymous bloggers. As before, I add abnormal returns or abnormal volumes for the 20 days preceding blog publications in the respective regressions, in order to control for momentum or reversals.

All tests for significance in this paper are discussed at the 10% significance level, unless otherwise stated. In tables, I distinguish between significance at the 1%, 5% and 10% level.

### **Data Sources**

I collect 500 buy and sell recommendations for the year 2006 from SeekingAlpha<sup>9</sup>; of those, 340 are classified as long recommendations and 160 as short<sup>10</sup>. The recommendations originate from 107 different blogs and are written by 122 distinct bloggers. SeekingAlpha is a website that provides links to over 200 financial blogs offering investment advice. It should be noted that SeekingAlpha does not accept submissions related to stock whose share price is inferior to \$1.00, as those are more likely to be manipulated. I consider that a desirable feature of the dataset, since results are more easily generalized to the “average” security. SeekingAlpha also offers brief biographies of the bloggers. A preliminary inspection of the profile of bloggers indicates extreme heterogeneity. Bloggers range from students, to people approaching investments as a hobby, to finance professionals. A large proliferation of institutional blogs – that is, blogs published by financial service providers – appears to indicate that professionals are

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<sup>8</sup> The “multiple recommendations” binary variable was excluded from the previous regression due to excessive collinearity with other predictors; as the correlation with other predictors appears non-significant in the controlled subset, it is included in this regression.

<sup>9</sup> <http://seekingalpha.com/>

<sup>10</sup> In forming the final sample, my raw dataset included all 310 short recommendations available during 2006 from Seeking Alpha (originating from 267 posts, some with multiple recommendations) and a randomly selected 660 long recommendations (originating from 500 randomly selected posts, out of total of 1,259 posts for the year 2006). It should be noted that I applied some filters when selecting blog recommendations; in particular, I only considered recommendations for common stock and excluded all posts that did not offer clear investment advice (that is, investment advice which could not be categorized as either a “buy” or “sell” recommendation). Similarly, I excluded all recommendations for which I could not obtain the original blog post. The final sample contains exactly 500 recommendations, of which 340 are long and 160 are short.

seeing blogs as a valid avenue to advertise their services and the performance of their recommendations.

In order to isolate the effects of blog recommendations from near simultaneous news announcements such as earnings releases, I search for news related to the firms mentioned in the advice and exclude all of the instances in which there were news related to a company in the interval between day -1 and day +1. In particular, I search for firm-specific news on the Factiva database originating from Dow-Jones Newswires or from the Wall Street Journal for the firms mentioned in the recommendations. I mark all observations not having any concurrent news as “controlled for news” and isolate those in a “controlled” subset. Similarly, I exclude from the “controlled” subset all recommendations which are close in date to other blog recommendations (that is, all recommendations that have other blog posts in the time interval between day -1 and day +1). The long-controlled subset contains 148 recommendations, while the short-controlled subset contains 60 recommendations. Overall, it should be noted that 208 out of 500 recommendations are given without concurrent news in the media or without other concurrent blog posts; this suggests that, while the majority of bloggers do no more than to capture and repackage information that is already available in other media, a portion of the blogs does spread novel information, be it in the form of analysis or news reporting. Out of the remaining 292 recommendations, in two instances I find that information was available on blogs prior to it being reported by either Dow Jones Newswires or The Wall Street Journal.

I obtain measures of Blog influence from Blogjuice.com<sup>11</sup>. The latter site offers various measures of blog audience. Of the available measures, I collect the Technorati Score<sup>12</sup> and the number of subscribers<sup>13</sup>.

I obtain daily trading volumes, market capitalization and closing prices adjusted for splits and dividend payments from the Wharton Center for Research in Security Prices (CRSP) database for the years 2005 and 2006. In order to investigate the future performance of stock recommendations published towards the end of 2006, I require the same information for 2007; I

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<sup>11</sup> [http://www.text-link-ads.com/blog\\_juice/](http://www.text-link-ads.com/blog_juice/)

<sup>12</sup> The Technorati score appears to be the most widely accepted measure of blog influence. It is compiled by Technorati, Inc. and it is available at: <http://www.technorati.com/>. Technorati, Inc. claims to be monitoring 103.8 million blogs and over 250 million WebPages; the score itself is based on a proprietary algorithms loosely based on the number of external links to a blog,

<sup>13</sup> Due to the high correlation between the Technorati score and the number of subscribers, the latter measure is dropped from the analysis, as the two appear to convey similar information. Neither variable is significant in cross-sectional regressions attempting to model blog accuracy or market reactions.

obtain the latter from Yahoo! Finance<sup>14</sup>. Data for the fitting of the three factor model used in the estimation of abnormal returns is obtained from Kenneth French's website<sup>15</sup>.

### **III. Analysis and Results**

#### **Descriptive Statistics**

Descriptive statistics offering a comparison of blogged securities versus all other securities are offered in *Table 1, Panel A*. Please note, the sample of "all other" securities includes all common stocks traded on NYSE, NASDAQ or AMEX. Mean and standard deviation figures reported are related to the two-year period spanning January 2005 to December 2006. Tests for significance are conducted as two-sample t-tests. The 500 recommendations span 346 distinct securities. Blogged securities appear to display higher mean daily returns (.07% versus .05%), higher mean trading volumes (circa 2.8 million shares daily versus circa 500 thousand shares daily and circa \$95 million in daily trading volume versus \$15 million) and higher average market capitalization (\$35 billion versus \$2billion).

*Panel B* pertains to the binary variables in the sample, reporting the number of observations and the proportion of observations coded as "1". Some interesting statistics are the fact that almost 45% of blog posts appear to have contemporaneous news reported by DowJones NewsWires and that almost 19% of blog posts are contemporaneous to other blog posts, indicating some time clustering<sup>16</sup>. *Panel C* offers descriptive statistics pertaining the distributions of the number of words in a blog post and of the Technorati Score of blogs. The mean number of words is circa 577, with a fairly wide range (the minimum is 9 and the maximum 3,670). The mean Technorati Score is 1,270,822, ranging from 0 to 8,386,150.

#### **Entire Sample – Event Studies**

Results for the event studies for the long and short subsets are presented, respectively, in *Table 2* and *Table 3*.

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<sup>14</sup> [www.yahoo.com](http://www.yahoo.com)

<sup>15</sup> <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>

<sup>16</sup> In order to control for time clustering, preliminary results for the event studies were computed also by forming portfolios of all blog recommendations published on the same day; while that led to some loss of power due to the lower number of observations, the main results proved robust to this alternative methodology.

*Table 2* reports that mean abnormal returns for the long sample are positive (circa 0.2%) on the event date, as expected, but lack statistical significance; abnormal returns are positive (circa 0.2%) and significant on day +1. Overall, the two-day return appears positive (circa 0.4%) and significant, but only circa 54% of the recommended securities display positive abnormal returns. Cumulative abnormal returns are negative (circa -0.8%) and significant over the 20 days preceding blog publication, providing evidence of bloggers being contrarian in their long picks. Cumulative returns are positive (circa 0.7%) and significant over the 20-days following blog publication, offering evidence in favor of blog accuracy<sup>17</sup>. Stronger evidence is offered by an analysis of abnormal trading volumes, as those appear positive for more than 50% of the firms on all days in the interval -6 to +10, while reaching the second-highest observed proportion on the event day, when 63.53% of the recommended stocks display positive abnormal trading volumes. Similarly, cumulative abnormal trading volumes appear positive and statistically significant on all examined intervals preceding and following the event day (20, 10 and 5 days).

*Table 3* reports that mean abnormal returns are negative and statistically significant on the event date, as expected, with a mean equal to -1.6%. Abnormal returns are negative and statistically significant also on day +1. The mean cumulative abnormal return for the 2-day event window is -1.84%. Also, over 76% of securities display negative cumulative 2-day abnormal returns. In addition, abnormal returns are negative and statistically significant on 3 out of 10 days preceding the blog publication (days -10, -7 and -1), which points to a tendency of bloggers to select stocks on the basis of a momentum strategy (stocks recommended for shorting appear to have performed poorly in the recent past). Consistently, I find negative returns for the 20, 10 and 5 days preceding the event day (but those cumulative returns are not significant). Cumulative abnormal returns are also negative but not significant on the 5, 10 and 20 day periods following the event date. Abnormal trading volumes appear positive and statistically significant on all examined intervals preceding and following the event day (20, 10 and 5 days).

Mean abnormal returns are plotted in *Figure 1*. The proportion of firms with positive abnormal trading volumes is plotted in *Figure 2*.

Results point to bloggers, on average, accurately predicting market movements; predictions appear more accurate in the short sample. The analysis of abnormal volumes points out to bloggers recommending securities displaying abnormal trading activity in the recent past, but with cumulative abnormal returns being negative during the preceding days in both the long

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<sup>17</sup> Since, as discussed below, blog recommendations do not appear to lead significant reactions for the 20-day event period (most of the reaction appears to take place on days 0 and day +1), the significant positive cumulative abnormal return is evidence not just of accuracy, but also of predictive ability of the blog.

and short sample, short recommendations appear to be momentum-based while long recommendations appear contrarian.

### **Entire Sample – ANOVA and Cross-Sectional Regressions.**

As previously discussed, abnormal returns can be interpreted as a measure of accuracy of blog recommendations. The analysis of variance that follows is aimed at determining whether blogs do differ in terms of quality of recommendations, by testing whether abnormal returns do vary by blog. The cross-sectional regressions attempt to identify which blog or blogger characteristics lead to superior recommendation quality.

Results for an analysis of variance of cumulative abnormal returns by blog are presented in *Table 4*. Cumulative abnormal returns for the 2, 5, 10 and 20 day event windows are computed by summing abnormal returns respectively over the intervals day 0 to day +1, day 0 to day +4, day 0 to day +9 and day 0 to day +19; cumulative abnormal returns for the short sample are multiplied by -1. The F-tests for differences in means indicate that blogs do differ in recommendation quality over the event day and the 2, 5 and 10 day event windows. Unreported results indicate that the most successful blog<sup>18</sup>, The Microcap Speculator<sup>19</sup>, obtains an average cumulative abnormal return of 9.96% over the 2-day event window (over 8 different recommendations); the least successful blog leads to negative 7.4% returns over the 2-day event window (but the sample includes only 2 different recommendations).

Having observed that blogs do differ in terms of recommendation quality, I attempt to identify which factors determine superior accuracy. In order to do so, in two separate procedures, I regress mean standardized cumulative abnormal returns<sup>20</sup> calculated over the 2 and 20 day event windows against a binary variable identifying short recommendations and against the previously discussed series of binary variables indicating blog and blogger characteristics; I also add, as a predictor, standardized cumulative abnormal returns for the 20 days interval preceding the blog publication date, in order to control for eventual momentum or reversals of returns. The estimated relation is:

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<sup>18</sup> Please note, only blogs with multiple recommendations are considered.

<sup>19</sup> <http://microcappeculator.net/>; The Microcap Speculator is published by an anonymous blogger and has been active since September 2005. As its name does suggest, the blog is oriented towards small capitalization stocks, often dealing with over-the-counter stocks (which have here been ignored). Somehow interestingly, all available measures of blog influence indicate that, despite its superior performance, the blog is below average in terms of visitors, subscribers and number of external links.

<sup>20</sup> As before, standardized abnormal returns for the short sample are multiplied by -1.

$$SCAR_i = \alpha_i + \beta_1 Short_i + \beta_2 Number\_Of\_Words_j + \beta_3 Blogger\_Pro_j + \beta_4 Blogger\_Student_j + \beta_5 Blogger\_Journalist_j + \beta_6 Blogger\_Certification_j + \beta_7 Blogger\_GradSchool_j + \beta_8 Blogger\_Institutional_k + \beta_9 Tech\_Score_k + \beta_{10} Investing_i + \beta_{11} SCAR\_PreEvent\_20days_i + \varepsilon_i$$

Where:

$SCAR_i$  is the standardized cumulative abnormal returns for recommendation  $i$  computed over two different time intervals: the 2-Day Event Window (day 0 and day +1) and the 20-day event window (day 0 to day +19);

$Short_i$  is a binary variable set at 1 if recommendation  $i$  is a short recommendation;

$Number\_of\_Words_i$  is the number of words in the blog post from which recommendation  $i$  originates;

$Blogger\_Pro_j$  is a binary variable set at 1 if blogger  $j$  is a financial professional or a retired financial professional;

$Blogger\_Student_j$  is a binary variable set at 1 if blogger  $j$  is a student of Finance or Economics;

$Blogger\_Journalist_j$  is a binary variable set at 1 if blogger  $j$  is or has been a journalist;

$Blogger\_Certification_j$  is a binary variable set at 1 if blogger  $j$  is a Certified Financial Analyst or if the blogger holds an equivalent foreign certification;

$Blogger\_GradSchool_j$  is a binary variable set at 1 if blogger  $j$  has received a graduate degree in Finance or Economics;

$Blogger\_Institutional_k$  is a binary variable set at 1 if blog  $k$  is published by or affiliated with an institution;

$Tech\_Score_k$  is the Technorati Score of blog  $k$ ;

$Investing_i$  is a binary variable set at 1 if the blog poster has a position invested in the security mentioned in recommendation  $i$ ;

$SCAR\_PreEvent\_20Days_i$  is the standardized cumulative abnormal return for the security mentioned in recommendation  $i$  over the 20 days preceding the recommendation of the same security.

$\varepsilon_i$  is an independent, identically distributed normal error term.

As indicated in *Table 5*, I find that bloggers are more accurate with short rather than long recommendations (with the result being significant over the 2-day event window but not over the 20-day event window). Bloggers with a graduate degree in Finance or Economics are more accurate in their predictions than bloggers without advanced degrees (but, again, the result is significant over the 2-day event window, not over the 20-day event window); in unreported regressions of cumulative abnormal returns on the same set of predictors, I estimate that recommendations by graduate students are about 1% higher over the 2-day event period. Finally, returns estimated over the preceding twenty days are significant and positively correlated, indicating that the accuracy of blogger recommendations is, at least in part, due to momentum<sup>21</sup>.

### **Controlled Sample – Event Studies**

Results for the event studies for the long and short controlled subsets are presented, respectively, in *Table 6* and *Table 7*.

*Table 6* reports that mean abnormal returns for the long controlled subset are positive on the event date (circa 0.2%), as expected, but lack statistical significance. Mean abnormal returns are positive and statistically significant on day +1 and for the two-day event period, averaging circa 0.5%. Positive and significant abnormal returns are observed also on days -2 and +10. Stronger evidence is offered by an analysis of abnormal trading volumes, as those appear positive and statistically significant on the event day, when 60.14% of the recommended stocks display positive abnormal trading volumes, and over the two-day event window (60.81% positive). Similarly, cumulative abnormal trading volumes appear positive and statistically significant over the 20 days preceding blog publication and over on all examined intervals following the event day (5, 10 and 20 days).

*Table 7* reports that, for the short-controlled sample, mean abnormal returns are negative and significant on the event date, with a mean equal to -0.8%. Two-day returns are negative but not statistically significant, but almost 67% of the recommended securities have negative 2-day returns (which is highly significant in a nonparametric sign test). Results are less clear when analyzing abnormal volumes. Individually, abnormal volumes appear positive for more than 50% of the examined securities on both the event day and on day +1, but results of a nonparametric sign test are not significant. On the other side, circa 63% of securities exhibit abnormal volumes

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<sup>21</sup> In order to make sure that multicollinearity was not biasing coefficient estimates, I attempt different model specification and find that results of cross-sectional regressions are robust to the insertion or removal of additional explanatory variables.

over the 2-day event window, which leads to a significant sign test. But an even higher proportion of firms displays positive abnormal volumes over the 20 days preceding the blog publication, questioning whether those abnormal volumes should really be interpreted as evidence for a market reaction to the blog itself. In addition, abnormal returns are negative and highly significant on day -1, giving once more voice to the idea that market movements are, at least in part, based on momentum.

Mean abnormal returns are plotted in *Figure 3*. The proportion of firms with positive abnormal trading volumes is plotted in *Figure 4*.

Overall, the results of the event studies of the controlled samples indicate that markets do react to blog posts; once more, this reaction appears asymmetrical, as short recommendations lead to a reaction that is about twice as large in magnitude (0.8% versus 0.5%).

### **Controlled Sample – ANOVA and Cross-Sectional Regressions.**

Having determined that blogs do lead to a market reaction, I investigate whether that reaction differs by blog and whether it is possible to identify factors which lead to stronger reactions. Accordingly, I employ an analysis of variance of cumulative abnormal returns and of abnormal trading volumes; results are presented in *Table 9*<sup>22</sup>. The F-tests for differences in means indicate that blogs do differ in terms of market reactions, both in terms of abnormal returns and abnormal trading volumes. Based on this analysis, the most influential blog appears to be The Microcap Speculator, with abnormal returns of circa 8.6% over the two-day event window (for 3 recommendations)<sup>23</sup>.

In order to identify which, if any, specific factors lead to stronger reactions, I regress mean standardized cumulative abnormal returns and abnormal trading volumes calculated over the 2-day event period against a binary variable identifying short recommendations, a series of binary variables indicating blog and blogger characteristics and proxying for depth of the analysis in the blog post and against market capitalization of the touted securities; I also add, as a predictor, standardized cumulative abnormal returns and cumulative abnormal volumes for the 20

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<sup>22</sup> As before, please note that abnormal returns and standardized abnormal returns in both the ANOVAs and regressions are multiplied by -1 for the short sample.

<sup>23</sup> The Microcap Speculator was identified as the most accurate blog in terms of predictions; while it is somehow reassuring to find out that the most accurate blog is also the most influential, the reader should be cautioned against putting excessive faith into those results, as the sample used to estimate reactions is a subset of the overall sample used to estimate accuracy of blog recommendations; in other words, both results could be driven by the same outlying observations. Further tests are possible but require a more extensive sample and are not of focal interest to the present discussion.

days interval preceding the blog publication date, in order to control for eventual momentum or reversals of returns and volumes. The hypothesized and tested regression relationships are then:

$$\begin{aligned} SCAR\_2days_i = & \alpha_i + \beta_1 Short_i + \beta_2 Multiple\_Recs_i + \beta_3 Number\_Of\_Words_i + \\ & \beta_4 Blogger\_Anonymous_j + \beta_5 Blogger\_Pro_j + \beta_6 Blogger\_Student_j + \\ & \beta_7 Blogger\_Journalist_j + \beta_8 Blogger\_Certification_j + \beta_9 Blogger\_GradSchool_j + \\ & \beta_{10} Blog\_Institutional_k + \beta_{11} Tech\_Score_k + \beta_{12} Investing_i + \\ & \beta_{13} Mean\_MCap_m + \beta_{14} SCAR\_PreEvent\_20Days_i + \varepsilon_i \end{aligned}$$

$$\begin{aligned} AbnVol\_2days_i = & \alpha_i + \beta_1 Short_i + \beta_2 Multiple\_Recs_i + \beta_3 Number\_Of\_Words_i + \\ & \beta_4 Blogger\_Anonymous_j + \beta_5 Blogger\_Pro_j + \beta_6 Blogger\_Student_j + \\ & \beta_7 Blogger\_Journalist_j + \beta_8 Blogger\_Certification_j + \beta_9 Blogger\_GradSchool_j + \\ & \beta_{10} Blog\_Institutional_k + \beta_{11} Tech\_Score_k + \beta_{12} Investing_i + \\ & \beta_{13} Mean\_MCap_m + \beta_{14} AbnVol\_PreEvent\_20Days_i + \varepsilon_i \end{aligned}$$

Where:

$Short_i$ ,  $Number\_Of\_Words_i$ ,  $Blogger\_Pro_j$ ,  $Blogger\_Student_j$ ,  $Blogger\_Journalist_j$ ,  $Blogger\_Certification_j$ ,  $Blogger\_GradSchool_j$ ,  $Blog\_Institutional_k$ ,  $Tech\_Score_k$ ,  $Investing_i$  and  $SCAR\_PreEvent\_20Days_i$  are defined as previously discussed;

$SCAR\_2days_i$  is the standardized cumulative abnormal returns for recommendation  $i$  computed over the 2-Day Event Window (day 0 and day +1);

$AbnVol\_2days_i$  is the abnormal trading volume for the security mentioned in recommendation  $i$  over the 2-Day Event Window (day 0 and day +1); daily abnormal trading volumes are calculated as the difference between daily share turnover and median share turnover; daily share turnover is calculated as the ratio of shares traded to shares outstanding, while the median is computed over the interval day -115 to day -21;

$MultipleRecs_i$  is a binary variable set at 1 if the blog post from which recommendation  $i$  originates contains multiple recommendations;

$Blogger\_Anonymous_j$  is a binary variable set at 1 if blogger  $j$  does not provide his/her first and last names;

$Mean\_MCap_m$  is the mean market capitalization for firm  $m$  over the period of time spanning January 2004 to December 2006;

$AbnVol\_PreEvent\_20days_i$  is the average abnormal trading volume for firm  $i$  over the 20 days preceding the blog publication day; daily abnormal trading volumes are calculated as the difference between daily share turnover and median share turnover; daily share turnover is calculated as the ratio of shares traded to shares outstanding, while the median is computed over the interval day -115 to day -21;

As reported in *Table 10*, I find that recommendations by graduate students lead to stronger reactions both in terms of abnormal returns and volumes; in unreported regressions of cumulative abnormal returns, I find that recommendations by graduate students lead on average to a 3% increase in the 2-day mean cumulative abnormal return. The positive coefficient on pre-event performance indicates that a portion of the abnormal returns is due to momentum, rather than to market reaction to the blog. Yet, even after controlling for momentum, the intercept for the regression of 2-day standardized cumulative abnormal returns is positive and significant, indicating that markets do react to blog recommendations. All other attempted predictors are not significant<sup>24</sup>.

#### **IV. Interpretation – Accuracy of Recommendations**

*Q1: How do bloggers select stocks to recommend?*

Recommended securities display higher returns, higher trading volumes and higher market capitalization than the “average” security<sup>25</sup>. There is strong evidence for abnormal trading volumes prior to blog publication. In all, bloggers appear to recommend large, highly visible securities; that is consistent with the findings of Wysocki (2001), as he reports that firms touted on stock message boards display high recent trading volume, and Dewally (2003), as he observes that advice given on message boards focuses on firms that have displayed recent abnormal performance. In short recommendations, bloggers prove to be momentum traders, as returns over the 20-days preceding the event appear negative (albeit not cumulatively significant). Negative

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<sup>24</sup> As before, in order to make sure that multicollinearity was not biasing coefficient estimates, I attempt different model specification and find that results of cross-sectional regressions are robust to the insertion or removal of additional explanatory variables.

<sup>25</sup> Means are computed over the two-year period spanning January 2005 to December 2006.

pre-event abnormal returns are observed in the long sample as well, indicating that, in selecting long stocks, bloggers behave as contrarians. Cross-sectional results confirm that accuracy of blog recommendations in both the short and long subsets is due, at least partially, to momentum.

*Q2: Are stock recommendations given on blogs a good source of investment advice?*

There is evidence for accuracy of both long and short recommendations; two-day abnormal returns are positive and significant for long recommendations (circa 0.4%), negative and statistically significant for the short sample (circa -1.8%). Bloggers appear to be more accurate in their short recommendations (with almost 66% of the securities displaying negative abnormal returns over the 2-day event window) than with positive recommendations (where the proportion of securities displaying positive abnormal returns is circa 54%)<sup>26</sup>. I fail to find an explanation as to why short recommendations appear more accurate, but the magnitude of the difference suggests that the gap is not fully explainable by differences in market reactions<sup>27</sup>. Cumulative abnormal returns over the 20 days following blog publication are positive and significant for the long sample and negative (but not significant) for the short sample.

*Q3: Do blogs differ in terms of quality of investment advice?*

Analysis of variance indicates that blogs do differ in terms of quality of investment advice. Cross-sectional analysis indicates that holders of graduate degrees predict market movements more accurately, but this reaction seems to be driven by observations from the “controlled” subset<sup>28</sup>. In other words, markets appear to react more to recommendations given by graduate degree holders,

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<sup>26</sup> Given that similar results have been observed for the “controlled” subset, I check for robustness, hypothesizing that both results could be driven by the same observations. Accordingly, I repeat the event study analysis on the subset of data having contemporaneous news, obtaining equivalent results.

<sup>27</sup> Note that accuracy of recommendation is the cumulative effect of predictive ability of the post and of market reaction to the post itself. While I do not formally discuss predictive ability in the present, for brevity, a comparison of abnormal returns of the overall and controlled samples indicates that, when news are present, abnormal returns are larger (the effect is more evident in the short subset, but it is observable in cumulative results for the long sample as well). I interpret the difference in price movements as evidence for predictive ability of blog recommendations.

<sup>28</sup> It should be noted that the binary variable identifying graduate degree holders is significant also in the regression for the subset controlled for news, which I interpret as evidence for markets reacting more to recommendations by graduate students; this raises the question whether graduate degree holders are truly predicting the markets more accurately or whether the observed abnormal returns are due to market reaction to the blog itself. In order to disentangle the two effects, I run the same regressions on a data subset containing only observations for which there are contemporaneous news. In this subset, the graduate degree binary variable carries a positive coefficient, but is not statistically significant. I interpret those results as indicating that, in this subset, the effect of news dwarfs the market reaction to the blog itself (as evidenced by the higher standard deviation of two day cumulative abnormal returns in the “with news” subset, circa 0.035 versus circa 0.02). I am forced to conclude that the higher returns associated with recommendations by graduate degrees holders are a consequence of market reaction to the blog itself, rather than evidence of superior predictive abilities.

but the evidence in favor of a superior predictive ability by graduate degree holders is, at best, weak.

### **Interpretation – Market Reaction**

*H1: Buy recommendations posted on blogs will lead to higher prices for the mentioned securities on the day of the posting and/or the following day; sell recommendations posted on blogs will lead to lower prices for the mentioned securities on the day of the posting and/or the following day.*

I find strong evidence for the above, as cumulative abnormal returns average a positive 0.5% (negative 0.8%) for long (short) recommendations over the two-day event window. Further evidence is given by the fact that, in the long sample, circa 54% of stocks display positive two-day abnormal returns; in the short sample, the proportion of negative abnormal returns is circa 66%.

*H2: Both buy and sell recommendations posted on blogs will lead to abnormally high trading volumes for the mentioned securities on the day of the posting and/or the following day.*

While I find statistically significant positive abnormal trading volumes during the 2-day event window, abnormal trading volumes are positive and significant during the preceding 20 days and abnormal trading volumes are strongly related to abnormal trading volumes observed in the 20 days preceding blog publication, suggesting the possibility that the observed results might be a result of trading momentum. Yet, a comparison of mean abnormal trading volumes by blog does indicate that some blogs do consistently lead to higher trading volume whenever posting a recommendations, providing support for the hypothesis.

*H3: Sell recommendations posted on blogs will lead to larger absolute-value abnormal returns and abnormal trading volumes on the day of the posting and/or the following day than buy recommendations.*

I find evidence in support of the above, as 2-day returns are about twice as large for short than for long recommendations (in absolute value, 0.8% versus 0.5%). Further evidence is given by the fact that, in the long sample, circa 54% of stocks display positive two-day abnormal returns, while, in the short sample, the proportion of negative abnormal returns is circa 66%. Yet, in the

cross-section, the binary variable identifying short recommendations appears insignificant, both in explaining abnormal returns and abnormal volumes. The stronger reaction to short recommendations is consistent with Groth et al. (1979) and Womack (1996) and is in contrast to findings by Liu and Smith (1990)<sup>29</sup>.

*H4: The impact on prices and trading volumes will be greater for stocks of firms with low market capitalizations.*

I find no evidence in support of the above.

*H5: The impact on prices and trading volumes will be greater for stocks touted on blogs written by individuals possessing one or more of the following credentials: being a finance professional or a retired finance professional, a finance student, a journalist, possessions of a Certified Financial Analyst designation or other equivalent financial certification, having earned a graduate degree in Finance or Economics.*

Of the hypothesized explanatory variables, only the binary variable identifying graduate degree holders appears significant. I estimate the effect of the graduate degree as leading to a 3% increase in 2-days abnormal returns and a 0.8% increase in share turnover.

*H6: The impact on prices and trading volumes will be weaker for stocks touted by anonymous bloggers.*

I find no evidence in support of the above. I observe that the most accurate and most influential blog in the sample is published by an anonymous blogger.

*H7: The impact on prices and trading volumes will be weaker for stocks touted by bloggers who do not hold a position in the touted security.*

I find no evidence in support of the above.

*H8: The impact on prices and trading volumes will be greater for stocks touted on blogs with a larger audience.*

I find no evidence in support of the above. Note that, necessarily, the testing of this hypothesis is actually a simultaneous test for the hypothesis that the Technorati score is a good proxy for size

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<sup>29</sup> While I find support for the stronger market reaction to short recommendations documented by Groth et al., I do not find evidence for the stronger impact of single-company recommendations documented in the same study. In all cross-sectional regressions, my binary variable identifying multiple recommendations is not significant.

of the audience. Rejection does not indicate whether market reaction is insensitive to size of the audience or whether the Technorati score is a poor proxy for audience size.

*H9: The impact on prices and trading volumes will be greater for stocks touted on institutional blogs.*

I find no evidence in support of the above. While coefficient estimates in the cross-sectional regressions are always positive, they are not significant.

*H10: The impact on prices and trading volumes will be greater for stocks touted on blog posts offering more in-depth analysis.*

I find no evidence in support of the above.

*H14: There will be no price reversion for the recommended stocks subsequent to the initial reaction.*

There appears to be no mean-reversion; while not always statistically significant, in both the short and long sample returns follow the momentum of the event day (cumulative abnormal returns over the 5, 10 and 20 days following the blog publication are positive in the long subset, negative in the short; cumulative abnormal returns are positive and significant over the 20 days following blog publication for the long sample and negative and significant over the 5 days following blog publication for the negative sample), reinforcing the initial reaction.

## **V. Conclusions**

I find that bloggers tend to recommend large, highly visible securities. I present evidence for abnormal trading volumes and for negative abnormal returns prior to the publication of blog recommendations. That is consistent with the findings of Wysocki (2001), as he reports that firms touted on stock message boards display high recent trading volume, and Dewally (2003), as he observes that advice given on message boards focuses on firms who have displayed recent abnormal performance. I also document that bloggers give momentum-based short recommendations but contrarian long recommendations; in contrast, Dewally finds that recommendations on message boards tend to be based on momentum.

There is evidence for accuracy of both long and short recommendations. For short recommendations the evidence is much stronger; shorting securities on the days they are given a

short recommendations would lead to an average return of 1.8% over the 2-day event window comprising the day of blog publication and the following trading day.

Blogs do differ in terms of quality of investment advice, but the present analysis does not support links between quality of investment advice and specific blog or blogger characteristics. There is evidence in support of higher abnormal returns associated with the blogger holding a graduate degree, but that appears to be due to market reaction to the blog itself, rather than to superior predictive abilities.

Blog recommendations do appear to have an impact on stock prices and trading volumes, which mirrors similar findings in regards to message boards by Antweiler and Frank (2004) and in regards to e-mail spam by Böhme and Holz (2006) and Frieder and Zittrain (2006). Average market reaction is of an increase (decrease) in stock price of circa 0.4% (0.8%) for long (short) recommendations. Abnormal trading volumes are positive as well, yet, trading volumes do appear abnormal in the days preceding the blog posting, which does leave open the possibility that abnormal trading volumes following blog publications are a result of trading momentum. The stronger impact of short recommendations<sup>30</sup> offers additional evidence in the still open debate as to whether short recommendations do lead to stronger market reactions; my results support Groth et al. (1979) and Womack (1996) but are in contrast with Liu and Smith (1990).

As hypothesized, I find no evidence of short-term mean reversion in the days following the event, contrary to what Böhme and Holz (2006) and Frieder and Zittrain (2006) find in their studies of the effects of e-mail spam. This difference in market reaction can be interpreted as an indication of the fact that, while e-mail spam is, often time, an attempt to manipulate markets, blogs are not. E-mail spammers spread false information in order to move the price of a security; eventually, the price reverts to its fundamental value. In the case of blogs, it is plausible to hypothesize that posters are acting out of a genuine desire to spread real information. The lack of mean reversion in prices following publication supports this hypothesis.

Bloggers do appear, for the most part, to echo information already available in the media. Yet, as market reactions to select blogs suggests, some do spread genuine information and analysis, which does lead to a market adjustment. Concerns about the potential for market manipulation are not motivated.

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<sup>30</sup> As previously noted, the stronger reaction to short recommendations, while evident in the event studies, is not supported by the cross-sectional analysis.

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## Tables

**TABLE 1 – Descriptive Statistics**

### Panel A

|                         | Blogged          |                 | All             |                 | t-test | p-value | Sig. |
|-------------------------|------------------|-----------------|-----------------|-----------------|--------|---------|------|
|                         | Mean             | Std             | Mean            | Std             |        |         |      |
| Mean Return             | 0.0007           | 0.0011          | 0.0005          | 0.0027          | 3.63   | 0.00    | ***  |
| Std. of Returns         | 0.0240           | 0.0124          | 0.0251          | 0.0182          | -1.60  | 0.11    |      |
| Mean # of Shares Traded | 2,893,808        | 6,687,882       | 515,126         | 2,266,119       | 6.60   | 0.00    | ***  |
| Mean Dollar Volume      | \$95,351,505     | \$240,554,768   | \$15,382,401    | \$120,782,478   | 6.15   | 0.00    | ***  |
| Mean Market Cap.        | \$34,657,824,743 | \$4,949,374,528 | \$1,895,710,347 | \$9,194,841,690 | 5.94   | 0.00    | ***  |

### Panel B

| Variable              | N   | Missing | Proportion Yes |
|-----------------------|-----|---------|----------------|
| Multiple_Recs         | 500 | 0       | 35.00%         |
| News_WSJ              | 485 | 15      | 20.82%         |
| News_DJ               | 485 | 15      | 44.54%         |
| Other_Blogs           | 500 | 0       | 18.80%         |
| Blogger_Anonymous     | 500 | 0       | 19.80%         |
| Blogger_Pro           | 500 | 0       | 36.60%         |
| Blogger_Student       | 500 | 0       | 8.00%          |
| Blogger_Journalist    | 500 | 0       | 20.60%         |
| Blogger_Certification | 500 | 0       | 20.40%         |
| Blogger_GradSchool    | 500 | 0       | 8.60%          |
| Blog_Institutional    | 498 | 2       | 33.94%         |
| Investing             | 500 | 0       | 35.40%         |

### Panel C

| Variable        | N   | Missing | Mean      | Median  | StDev     | Max       | Min |
|-----------------|-----|---------|-----------|---------|-----------|-----------|-----|
| Number_Of_Words | 499 | 1       | 577.10    | 270.50  | 447.60    | 3,670.00  | 9   |
| Tech            | 465 | 35      | 1,270,822 | 403,983 | 2,045,589 | 8,396,150 | 0   |

*Panel A* reports mean and standard deviation for daily mean returns, daily return volatility (computed as the standard deviation of returns), daily mean trading volume (number of shares and dollars) and mean market capitalization for blogged securities and for all common stock traded on the NYSE, NASDAQ and AMEX and results for two-sample t-tests for differences in means. All statistics are calculated by security, over the 2-year interval starting January 2005 and ending December 2006. Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05” level; “\*\*\*” indicates significance at the .01 level.

*Panel B* reports the number of observations, the number of missing observations and the percent of “Yes” for the binary variables in the sample: *Multiple\_Recs* (“Yes” if the post contains recommendations concerning multiple securities), *News\_WSJ* and *News\_DJ* (“Yes” if news regarding the securities have been reported on days -1, 0 or +1 on, respectively, The Wall Street Journal or on DowJones NewsWire), *Other\_Blogs* (“Yes” if the security has been recommended on other blogs on days -1, 0 or +1), *Blogger\_Anonymous* (“Yes” if the blogger does not reveal both his first and last names), *Blogger\_Pro* (“Yes” if the blogger is a finance professional or a retired finance professional), *Blogger\_Student* (“Yes” if the blogger is a student of Finance or Economics), *Blogger\_Journalist* (“Yes” if the blogger is or has been a journalist; please note, publication of a blog does not qualify), *Blogger\_Certification* (if the blogger is a Certified Financial Analyst or holds an equivalent foreign certification), *Blogger\_GradSchool* (“Yes” if the blogger has earned a graduate or doctoral degree in Finance or Economics; please note, current studies do not qualify), *Blog\_Institutional* (“Yes” if the blog is published by or affiliated with a commercial enterprise) and *Investing* (“Yes” if the blogger or his/her relatives or clients have invested in the security or in funds holding the security).

*Panel C* reports the number of observations, the number of missing observations, the mean, the median, the standard deviation, the maximum and the minimum for the variables *Number\_Of\_Words* (the number of words in the blog post) and *Tech* (the Technorati score for the blog).

**TABLE 2 – Event Study, Long Sample**

| Relative Date      | Mean<br>AR/CAR | <i>p1</i>       | Abn<br>Ret. >0 | <i>p2</i>   | Abn.<br>Volume<br>> 0 | <i>p3</i>           |
|--------------------|----------------|-----------------|----------------|-------------|-----------------------|---------------------|
| -10                | -0.0006        | 0.44            | 48.82%         | 0.66        | 51.76%                | 0.52                |
| -9                 | -0.0002        | 0.95            | 50.29%         | 0.91        | 47.06%                | 0.28                |
| -8                 | 0.0002         | 0.84            | 49.12%         | 0.74        | 49.12%                | 0.74                |
| -7                 | 0.0007         | 0.97            | 48.53%         | 0.59        | 48.53%                | 0.59                |
| -6                 | -0.0002        | 0.80            | 47.94%         | 0.45        | 52.06%                | 0.45                |
| -5                 | -0.0009        | 0.55            | 47.65%         | 0.39        | 50.00%                | 1.00                |
| -4                 | 0.0009         | 0.91            | 48.24%         | 0.52        | 55.00%                | 0.07 *              |
| -3                 | 0.0000         | 0.95            | 47.94%         | 0.45        | 56.76%                | 0.01 **             |
| -2                 | 0.0007         | 0.21            | 48.82%         | 0.66        | 55.88%                | 0.03 **             |
| -1                 | -0.0014        | 0.08 *          | 51.47%         | 0.59        | 58.53%                | <0.01 ***           |
| <b>0</b>           | <b>0.0020</b>  | <b>0.38</b>     | <b>52.65%</b>  | <b>0.33</b> | <b>63.53%</b>         | <b>&lt;0.01</b> *** |
| <b>1</b>           | <b>0.0020</b>  | <b>0.04</b> **  | <b>52.35%</b>  | <b>0.39</b> | <b>61.47%</b>         | <b>&lt;0.01</b> *** |
| 2                  | 0.0000         | 0.93            | 52.06%         | 0.45        | 63.53%                | <0.01 ***           |
| 3                  | -0.0027        | 0.01 **         | 45.00%         | 0.07 *      | 54.12%                | 0.13                |
| 4                  | 0.0008         | 0.42            | 47.35%         | 0.33        | 56.76%                | 0.01 **             |
| 5                  | -0.0004        | 0.57            | 47.35%         | 0.33        | 63.82%                | 0.00 ***            |
| 6                  | -0.0017        | 0.16            | 42.35%         | 0.00 ***    | 56.18%                | 0.02 **             |
| 7                  | -0.0008        | 0.63            | 49.12%         | 0.74        | 58.82%                | <0.01 ***           |
| 8                  | 0.0016         | 0.15            | 47.35%         | 0.33        | 53.53%                | 0.19                |
| 9                  | 0.0027         | 0.00 ***        | 45.29%         | 0.08 *      | 56.47%                | 0.02 **             |
| 10                 | 0.0016         | 0.10 *          | 49.12%         | 0.74        | 57.65%                | <0.01 ***           |
| 20 Days Pre-Event  | -0.0085        | 0.06 **         | 46.47%         | 0.19        | 67.94%                | <0.01 ***           |
| 10 Days Pre-Event  | -0.0009        | 0.51            | 52.94%         | 0.28        | 63.53%                | <0.01 ***           |
| 5 Days Pre-Event   | -0.0007        | 0.67            | 49.71%         | 0.91        | 64.12%                | <0.01 ***           |
| <b>2-Day Event</b> | <b>0.0039</b>  | <b>0.03</b> *** | <b>53.82%</b>  | <b>0.16</b> | <b>65.88%</b>         | <b>&lt;0.01</b> *** |
| 5-Day Event        | 0.0021         | 0.57            | 50.59%         | 0.83        | 66.18%                | <0.01 ***           |
| 10-Day Event       | 0.0035         | 0.17            | 51.18%         | 0.66        | 68.24%                | <0.01 ***           |
| 20-Day Event       | 0.0068         | 0.10 *          | 55.88%         | 0.03 ***    | 68.53%                | <0.01 ***           |

Event Study results for “Long Recommendations”. *Relative Date* indicates the date or time period of interest, relative to the date of blog publication (date 0). “20 Days Pre-Event”, “10 Days Pre-Event” and “5 Days Pre-Event” indicate cumulative results for the periods spanning, respectively, day -20 to day -1, day -10 to day -1 and day -5 to day -1. “2-day Event” indicates cumulative results for day 0 and day +1. “5-Day Event”, “10-Day Event” and “20-Day Event” indicate cumulative results for the periods spanning, respectively, day 0 to day +4, day 0 to day +9 and day 0 to day +19. *Mean AR/CAR* reports mean abnormal returns for each of the days of interest or mean cumulative abnormal returns for the sub-periods of interest. *p1* is the p-value of a test for significance of abnormal returns (or cumulative abnormal returns) based on standardized abnormal returns, equivalent to the p-value associated with the *J2* statistic as described by Campbell, Lo and Mackinlay (1997). *Abn. Ret. > 0* reports the proportion of firms for which abnormal returns are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Ret. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal returns is 50%. *Abn. Vol. > 0* reports the proportion of firms for which abnormal trading volumes (defined as the difference between share turnover and median share turnover) are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Vol. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal volumes is 50%. Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 3 – Event Study, Short Sample**

| Relative Date      | Mean<br>AR/CAR | <i>p1</i>           | Abn<br>Ret. >0 | <i>p2</i>           | Abn.<br>Volume<br>> 0 | <i>p3</i>           |
|--------------------|----------------|---------------------|----------------|---------------------|-----------------------|---------------------|
| -10                | -0.0019        | 0.03 **             | 48.75%         | 0.75                | 53.75%                | 0.34                |
| -9                 | 0.0012         | 0.28                | 55.63%         | 0.15                | 51.88%                | 0.64                |
| -8                 | 0.0043         | 0.72                | 47.50%         | 0.53                | 56.25%                | 0.11                |
| -7                 | -0.0031        | 0.04 **             | 51.25%         | 0.75                | 51.25%                | 0.75                |
| -6                 | -0.0002        | 0.91                | 45.63%         | 0.27                | 56.25%                | 0.11                |
| -5                 | -0.0013        | 0.58                | 49.38%         | 0.87                | 61.25%                | <0.01 ***           |
| -4                 | 0.0045         | <0.01 ***           | 53.75%         | 0.34                | 61.25%                | <0.01 ***           |
| -3                 | -0.0007        | 0.64                | 45.63%         | 0.27                | 61.88%                | <0.01 ***           |
| -2                 | 0.0024         | 0.31                | 48.75%         | 0.75                | 66.88%                | <0.01 ***           |
| -1                 | -0.0087        | 0.01 ***            | 47.50%         | 0.53                | 70.63%                | <0.01 ***           |
| <b>0</b>           | <b>-0.0161</b> | <b>&lt;0.01</b> *** | <b>35.63%</b>  | <b>&lt;0.01</b> *** | <b>60.00%</b>         | <b>0.01</b> **      |
| <b>1</b>           | <b>-0.0024</b> | <b>0.03</b> **      | <b>37.50%</b>  | <b>&lt;0.01</b> *** | <b>65.00%</b>         | <b>&lt;0.01</b> *** |
| 2                  | 0.0006         | 0.59                | 46.25%         | 0.34                | 57.50%                | 0.06 *              |
| 3                  | -0.0014        | 0.58                | 48.75%         | 0.75                | 63.13%                | <0.01 ***           |
| 4                  | -0.0024        | 0.88                | 47.50%         | 0.53                | 60.00%                | 0.01 **             |
| 5                  | 0.0019         | 0.14                | 43.13%         | 0.08 *              | 59.38%                | 0.02 **             |
| 6                  | 0.0030         | 0.12                | 57.50%         | 0.06 *              | 60.00%                | 0.01 **             |
| 7                  | 0.0059         | 0.01 **             | 54.38%         | 0.27                | 51.88%                | 0.64                |
| 8                  | -0.0010        | 0.63                | 46.25%         | 0.34                | 50.00%                | 1.00                |
| 9                  | -0.0021        | 0.34                | 49.38%         | 0.87                | 51.88%                | 0.64                |
| 10                 | -0.0017        | 0.18                | 44.38%         | 0.15                | 47.50%                | 0.53                |
| 20 Days Pre-Event  | -0.0009        | 0.33                | 52.50%         | 0.53                | 75.63%                | <0.01 ***           |
| 10 Days Pre-Event  | -0.0035        | 0.14                | 48.75%         | 0.75                | 71.88%                | <0.01 ***           |
| 5 Days Pre-Event   | -0.0038        | 0.92                | 50.63%         | 0.87                | 71.88%                | <0.01 ***           |
| <b>2-Day Event</b> | <b>-0.0184</b> | <b>&lt;0.01</b> *** | <b>33.13%</b>  | <b>&lt;0.01</b> *** | <b>68.13%</b>         | <b>&lt;0.01</b> *** |
| 5-Day Event        | -0.0216        | <0.01 ***           | 40.00%         | 0.01 ***            | 70.63%                | <0.01 ***           |
| 10-Day Event       | -0.0140        | 0.41                | 51.88%         | 0.64                | 69.38%                | <0.01 ***           |
| 20-Day Event       | -0.0324        | 0.11                | 49.38%         | 0.87                | 72.50%                | <0.01 ***           |

Event Study results for “Short Recommendations”. *Relative Date* indicates the date or time period of interest, relative to the date of blog publication (date 0). “20 Days Pre-Event”, “10 Days Pre-Event” and “5 Days Pre-Event” indicate cumulative results for the periods spanning, respectively, day -20 to day -1, day -10 to day -1 and day -5 to day -1. “2-day Event” indicates cumulative results for day 0 and day +1. “5-Day Event”, “10-Day Event” and “20-Day Event” indicate cumulative results for the periods spanning, respectively, day 0 to day +4, day 0 to day +9 and day 0 to day +19. *Mean AR/CAR* reports mean abnormal returns for each of the days of interest or mean cumulative abnormal returns for the sub-periods of interest. *p1* is the p-value of a test for significance of abnormal returns (or cumulative abnormal returns) based on standardized abnormal returns, equivalent to the p-value associated with the *J2* statistic as described by Campbell, Lo and Mackinlay (1997). *Abn. Ret. > 0* reports the proportion of firms for which abnormal returns are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Ret. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal returns is 50%. *Abn. Vol. > 0* reports the proportion of firms for which abnormal trading volumes (defined as the difference between share turnover and median share turnover) are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Vol. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal volumes is 50%. Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 4 – ANOVA, Cumulative Abnormal Returns by blog**

| Timeframe           | F-stat | p-value | Significance |
|---------------------|--------|---------|--------------|
| Event Day           | 2.34   | < 0.001 | ***          |
| 2-Day Event Window  | 1.87   | < 0.001 | ***          |
| 5-Day Event Window  | 1.96   | < 0.001 | ***          |
| 10-Day Event Window | 1.37   | 0.046   | **           |
| 20-Day Event Window | 1.17   | 0.196   |              |

Results of an analysis of variance for cumulative abnormal returns by blog. Please note that cumulative abnormal returns for the short subset have been multiplied by -1. The tests compare mean cumulative abnormal returns by blog; the null hypothesis is that mean cumulative abnormal returns for all blogs are equal. When conducting the tests, all blogs for which only one observation is available are grouped into one single group; the sample contains 25 blogs with only one observation and 57 blogs with multiple observations (ranging from 2 to 35 observations each).

*Timeframe* indicates the time period of interest, relative to the date of blog publication (the “Event Day” or date 0). “2-day Event” indicates cumulative results for day 0 and day +1. “5-Day Event”, “10-Day Event” and “20-Day Event” indicate cumulative results for the periods spanning, respectively, day 0 to day +4, day 0 to day +9 and day 0 to day +19.

Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 5 – Cross-Sectional Regressions, Standardized Cumulative Abnormal Returns**

| Predictor                    | SCARs              |         |     |                     |         |     |
|------------------------------|--------------------|---------|-----|---------------------|---------|-----|
|                              | 2-Day Event Window |         |     | 20-Day Event Window |         |     |
|                              | Coef               | p-value | Sig | Coef                | p-value | Sig |
| Constant                     | 0.1740             | 0.40    |     | 0.1010              | 0.53    |     |
| Short                        | 0.3539             | 0.04    | **  | 0.0331              | 0.81    |     |
| Number_Of_Words <sup>1</sup> | 0.6500             | 0.97    |     | 14.6000             | 0.32    |     |
| Blogger_Pro                  | -0.2858            | 0.18    |     | -0.1336             | 0.43    |     |
| Blogger_Student              | -0.4227            | 0.27    |     | -0.4082             | 0.18    |     |
| Blogger_Journalist           | -0.1363            | 0.58    |     | -0.1525             | 0.43    |     |
| Blogger_Certification        | 0.1110             | 0.68    |     | 0.1122              | 0.59    |     |
| Blogger_GradSchool           | 0.5327             | 0.09    | *   | 0.1914              | 0.44    |     |
| Blog_Institutional           | 0.3430             | 0.13    |     | 0.1862              | 0.29    |     |
| Tech_Score <sup>1</sup>      | -0.0030            | 0.51    |     | 0.0000              | 0.98    |     |
| Investing                    | 0.0085             | 0.96    |     | -0.0632             | 0.65    |     |
| SCAR Pre-Event 20 days       | 0.1237             | 0.03    | **  | 0.0929              | 0.04    | **  |
| <b>R<sup>2</sup></b>         | 4.5%               |         |     | 2.3%                |         |     |
| <b>N</b>                     | 462                |         |     | 462                 |         |     |

<sup>1</sup> Coefficient Estimates are multiplied by 10<sup>5</sup>

Cross-sectional regressions of standardized cumulative abnormal returns versus the hypothesized predictors (please note that SCARs for short recommendations are multiplied by -1). Dates are relative to the date of blog publication (the “Event Day” or date 0); hence, the “2-day Event Window” indicates cumulative results for day 0 and day +1 and the “20-Day Event Window” indicates cumulative results for the periods spanning day 0 to day +19. The table reports coefficient estimates, p-values associated with *t*-test for significance and overall *R-squared* statistics for the regression:

$$\begin{aligned}
 SCAR_i = & \alpha_i + \beta_1 Short_i + \beta_2 Number\_Of\_Words_i + \beta_3 Blogger\_Pro_j + \\
 & \beta_4 Blogger\_Student_j + \beta_5 Blogger\_Journalist_j + \beta_6 Blogger\_Certification_j + \\
 & \beta_7 Blogger\_GradSchool_j + \beta_8 Blogger\_Institutional_k + \beta_9 Tech\_Score_k + \beta_{10} Investing_i + \\
 & \beta_{11} SCAR\_PreEvent\_20days_i + \varepsilon_i
 \end{aligned}$$

Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 6 – Event Study, Long-Controlled Sample**

| Relative Date      | Mean<br>AR/CAR | <i>p1</i>   | Abn<br>Ret. >0 | <i>p2</i>   | Abn.<br>Volume<br>> 0 | <i>p3</i>   |            |
|--------------------|----------------|-------------|----------------|-------------|-----------------------|-------------|------------|
| -10                | -0.0004        | 0.81        | 50.00%         | 1.00        | 43.92%                | 0.14        |            |
| -9                 | -0.0010        | 0.47        | 45.95%         | 0.32        | 40.54%                | 0.02        | **         |
| -8                 | 0.0012         | 0.81        | 50.68%         | 0.87        | 48.65%                | 0.74        |            |
| -7                 | 0.0020         | 0.15        | 52.70%         | 0.51        | 44.59%                | 0.19        |            |
| -6                 | -0.0023        | 0.44        | 46.62%         | 0.41        | 45.95%                | 0.32        |            |
| -5                 | -0.0031        | 0.18        | 48.65%         | 0.74        | 45.95%                | 0.32        |            |
| -4                 | 0.0015         | 0.50        | 50.68%         | 0.87        | 45.27%                | 0.25        |            |
| -3                 | 0.0002         | 0.71        | 51.35%         | 0.74        | 54.05%                | 0.32        |            |
| -2                 | 0.0037         | 0.02        | 54.73%         | 0.25        | 53.38%                | 0.41        | **         |
| -1                 | -0.0020        | 0.27        | 48.65%         | 0.74        | 48.65%                | 0.74        |            |
| <b>0</b>           | <b>0.0016</b>  | <b>0.23</b> | <b>52.70%</b>  | <b>0.51</b> | <b>60.14%</b>         | <b>0.01</b> | <b>**</b>  |
| <b>1</b>           | <b>0.0031</b>  | <b>0.06</b> | <b>54.05%</b>  | <b>0.32</b> | <b>54.73%</b>         | <b>0.25</b> | <b>*</b>   |
| 2                  | 0.0013         | 0.33        | 56.08%         | 0.14        | 60.81%                | 0.01        | ***        |
| 3                  | 0.0001         | 0.96        | 48.65%         | 0.74        | 47.30%                | 0.51        |            |
| 4                  | -0.0013        | 0.86        | 45.95%         | 0.32        | 47.97%                | 0.62        |            |
| 5                  | 0.0001         | 0.85        | 48.65%         | 0.74        | 63.51%                | <0.01       | ***        |
| 6                  | -0.0001        | 0.79        | 41.89%         | 0.05        | 51.35%                | 0.74        | **         |
| 7                  | -0.0016        | 0.99        | 43.92%         | 0.14        | 53.38%                | 0.41        |            |
| 8                  | 0.0024         | 0.18        | 47.30%         | 0.51        | 52.03%                | 0.62        |            |
| 9                  | 0.0007         | 0.85        | 42.57%         | 0.07        | 54.05%                | 0.32        | *          |
| 10                 | 0.0045         | 0.06        | 42.57%         | 0.07        | 54.05%                | 0.32        | *          |
| 20 Days Pre-Event  | -0.0112        | 0.26        | 45.27%         | 0.25        | 63.51%                | <0.01       | ***        |
| 10 Days Pre-Event  | -0.0002        | 0.91        | 52.70%         | 0.51        | 54.73%                | 0.25        |            |
| 5 Days Pre-Event   | 0.0002         | 0.69        | 50.68%         | 0.87        | 56.76%                | 0.10        |            |
| <b>2-Day Event</b> | <b>0.0047</b>  | <b>0.03</b> | <b>54.73%</b>  | <b>0.25</b> | <b>60.81%</b>         | <b>0.01</b> | <b>***</b> |
| 5 Days Post-Event  | 0.0001         | 0.74        | 51.35%         | 0.74        | 60.14%                | 0.01        | **         |
| 10 Days Post-Event | 0.0054         | 0.32        | 53.38%         | 0.41        | 60.81%                | 0.01        | ***        |
| 20 Days Post-Event | 0.0009         | 0.78        | 47.30%         | 0.51        | 64.19%                | <0.01       | ***        |

Event Study results for “Long-Controlled Recommendations”. *Relative Date* indicates the date or time period of interest, relative to the date of blog publication (date 0). “20 Days Pre-Event”, “10 Days Pre-Event” and “5 Days Pre-Event” indicate cumulative results for the periods spanning, respectively, day -20 to day -1, day -10 to day -1 and day -5 to day -1. “2-day Event” indicates cumulative results for day 0 and day +1. “5 Days Post-Event”, “10 Days Post-Event” and “20 Days Post-Event” indicate cumulative results for the periods spanning, respectively, day 2 to day +6, day 2 to day +11 and day 2 to day +21. *Mean AR/CAR* reports mean abnormal returns for each of the days of interest or mean cumulative abnormal returns for the sub-periods of interest. *p1* is the p-value of a test for significance of abnormal returns or cumulative abnormal returns based on standardized abnormal returns, equivalent to the p-value associated with the *J2* statistic as described by Campbell, Lo and Mackinlay (1997). *Abn. Ret. > 0* reports the proportion of firms for which abnormal returns are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Ret. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal returns is 50%. *Abn. Vol. > 0* reports the proportion of firms for which abnormal volumes (defined as the difference between share turnover and median share turnover) are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Vol. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal volumes is 50%. Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 7 – Event Study, Short-Controlled Sample**

| Relative Date      | Mean<br>AR/CAR | <i>p1</i>     | Abn<br>Ret. >0 | <i>p2</i>       | Abn.<br>Volume<br>> 0 | <i>p3</i>      |
|--------------------|----------------|---------------|----------------|-----------------|-----------------------|----------------|
| -10                | -0.0042        | 0.02 **       | 51.67%         | 0.80            | 56.67%                | 0.30           |
| -9                 | 0.0014         | 0.63          | 51.67%         | 0.80            | 51.67%                | 0.80           |
| -8                 | -0.0011        | 0.90          | 46.67%         | 0.61            | 51.67%                | 0.80           |
| -7                 | 0.0039         | 0.23          | 61.67%         | 0.07 *          | 46.67%                | 0.61           |
| -6                 | -0.0007        | 0.63          | 41.67%         | 0.20            | 56.67%                | 0.30           |
| -5                 | -0.0018        | 0.84          | 50.00%         | 1.00            | 63.33%                | 0.04 **        |
| -4                 | 0.0048         | 0.03 **       | 55.00%         | 0.44            | 55.00%                | 0.44           |
| -3                 | -0.0060        | 0.09 *        | 38.33%         | 0.07 *          | 60.00%                | 0.12           |
| -2                 | 0.0006         | 0.56          | 40.00%         | 0.12            | 66.67%                | 0.01 ***       |
| -1                 | -0.0201        | <0.01 ***     | 48.33%         | 0.80            | 61.67%                | 0.07 *         |
| <b>0</b>           | <b>-0.0084</b> | <b>0.07 *</b> | <b>36.67%</b>  | <b>0.04 **</b>  | <b>53.33%</b>         | <b>0.61</b>    |
| <b>1</b>           | <b>0.0004</b>  | <b>0.80</b>   | <b>41.67%</b>  | <b>0.20</b>     | <b>58.33%</b>         | <b>0.20</b>    |
| 2                  | 0.0032         | 0.07 *        | 56.67%         | 0.30            | 60.00%                | 0.12           |
| 3                  | -0.0050        | 0.24          | 38.33%         | 0.07 *          | 61.67%                | 0.07 *         |
| 4                  | -0.0017        | 0.50          | 45.00%         | 0.44            | 50.00%                | 1.00           |
| 5                  | -0.0028        | 0.66          | 36.67%         | 0.04 **         | 56.67%                | 0.30           |
| 6                  | 0.0012         | 0.28          | 55.00%         | 0.44            | 60.00%                | 0.12           |
| 7                  | 0.0053         | 0.21          | 55.00%         | 0.44            | 51.67%                | 0.80           |
| 8                  | 0.0004         | 0.90          | 51.67%         | 0.80            | 43.33%                | 0.30           |
| 9                  | -0.0027        | 0.84          | 43.33%         | 0.30            | 46.67%                | 0.61           |
| 10                 | -0.0014        | 0.42          | 43.33%         | 0.30            | 38.33%                | 0.07 *         |
| 20 Days Pre-Event  | -0.0233        | 0.57          | 51.67%         | 0.80            | 78.33%                | <0.01 ***      |
| 10 Days Pre-Event  | -0.0232        | 0.36          | 43.33%         | 0.30            | 73.33%                | <0.01 ***      |
| 5 Days Pre-Event   | -0.0224        | 0.23          | 41.67%         | 0.20            | 71.67%                | <0.01 ***      |
| <b>2-Day Event</b> | <b>-0.0080</b> | <b>0.27</b>   | <b>33.33%</b>  | <b>0.01 ***</b> | <b>63.33%</b>         | <b>0.04 **</b> |
| 5 Days Post-Event  | -0.0052        | 0.39          | 45.00%         | 0.44            | 63.33%                | 0.04 **        |
| 10 Days Post-Event | -0.0037        | 0.57          | 53.33%         | 0.61            | 61.67%                | 0.07 *         |
| 20 Days Post-Event | -0.0293        | 0.45          | 48.33%         | 0.80            | 63.33%                | 0.04 **        |

Event Study results for “Short-Controlled Recommendations”. *Relative Date* indicates the date or time period of interest, relative to the date of blog publication (date 0). “20 Days Pre-Event”, “10 Days Pre-Event” and “5 Days Pre-Event” indicate cumulative results for the periods spanning, respectively, day -20 to day -1, day -10 to day -1 and day -5 to day -1. “2-day Event” indicates cumulative results for day 0 and day +1. “5 Days Post-Event”, “10 Days Post-Event” and “20 Days Post-Event” indicate cumulative results for the periods spanning, respectively, day 2 to day +6, day 2 to day +11 and day 2 to day +21. *Mean AR/CAR* reports mean abnormal returns for each of the days of interest or mean cumulative abnormal returns for the sub-periods of interest. *p1* is the p-value of a test for significance of abnormal returns or cumulative abnormal returns based on standardized abnormal returns, equivalent to the p-value associated with the *J2* statistic as described by Campbell, Lo and Mackinlay (1997). *Abn. Ret. > 0* reports the proportion of firms for which abnormal returns are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Ret. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal returns is 50%. *Abn. Vol. > 0* reports the proportion of firms for which abnormal volumes (defined as the difference between share turnover and median share turnover) are positive during the period of interest. *p3* is the p-value of a non-parametric sign test on *Abn. Vol. > 0*; the null hypothesis is that the true proportion of firms with positive abnormal volumes is 50%. Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 8 – ANOVA, Standardized Cumulative Abnormal Returns and Abnormal Trading Volumes, Long-Controlled Sample**

***Panel A – Cumulative Abnormal Returns***

| Timeframe          | F-stat | p-value | Significance |
|--------------------|--------|---------|--------------|
| Event Day          | 1.91   | 0.003   | ***          |
| 2 Day Event Window | 1.6    | 0.023   | **           |
| 5 Days Post-Event  | 0.92   | 0.601   |              |
| 10 Days Post-Event | 1.14   | 0.285   |              |
| 20 Days Post-Event | 1.75   | 0.009   | ***          |

***Panel B – Abnormal Trading Volumes***

| Timeframe          | F-stat | p-value | Significance |
|--------------------|--------|---------|--------------|
| Event Day          | 2.02   | 0.001   | ***          |
| 2 Day Event Window | 1.43   | 0.065   | *            |
| 5 Days Post-Event  | 2.76   | < 0.001 | ***          |
| 10 Days Post-Event | 2.86   | < 0.001 | ***          |
| 20 Days Post-Event | 2.93   | < 0.001 | ***          |

*Panel A:* Results of an analysis of variance for cumulative abnormal returns by blog. Please note that cumulative abnormal returns for the short subset have been multiplied by -1. The tests compare mean cumulative abnormal returns by blog; the null hypothesis is that mean cumulative abnormal returns for all blogs are equal.

*Panel A:* Results of an analysis of variance for abnormal trading volumes by blog. Abnormal trading volumes are estimated as the difference between daily share turnover and median daily share turnover for each security (where share turnover is estimated as the ratio of traded shares to shares outstanding and the median is computed over the period spanning day -115 to day -21. The tests compare mean abnormal volumes by blog; the null hypothesis is that abnormal trading volumes for all blogs are equal.

In both panels, when conducting the tests, all blogs for which only one observation is available are grouped into one single group; the sample contains 20 blogs with only one observation and 37 blogs with multiple observations (ranging from 2 to 15 observations each). *Timeframe* indicates the date or time period of interest, relative to the date of blog publication (“Event Day” or date 0). “2-day Event” indicates cumulative results for day 0 and day +1. “5 Days Post-Event”, “10 Days Post-Event” and “20 Days Post-Event” indicate cumulative results for the periods spanning, respectively, day +2 to day +6, day +2 to day +11 and day +2 to day +21. Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*” indicates significance at the .05 level; “\*\*\*” indicates significance at the .01 level.

**TABLE 9 – Cross-Sectional Regressions, Controlled Sample**

| Controlled Sample            |             |         |     |                       |         |     |
|------------------------------|-------------|---------|-----|-----------------------|---------|-----|
| Predictor                    | 2-Day SCARs |         |     | 2-Day Abnormal Volume |         |     |
|                              | Coef        | p-value | Sig | Coef                  | p-value | Sig |
| Constant                     | 0.6120      | 0.01    | *** | 0.0031                | 0.37    |     |
| Short                        | -0.1893     | 0.31    |     | 0.0015                | 0.56    |     |
| Multiple_Recs                | -0.1544     | 0.40    |     | -0.0026               | 0.32    |     |
| Number_Of_Words <sup>1</sup> | -18.1900    | 0.29    |     | -0.3670               | 0.12    |     |
| Blogger_Anonymous            | -0.0119     | 0.96    |     | 0.0019                | 0.52    |     |
| Blogger_Pro                  | 0.2826      | 0.18    |     | -0.0265               | 0.36    |     |
| Blogger_Student              | -0.1992     | 0.62    |     | -0.0041               | 0.47    |     |
| Blogger_Journalist           | -0.3144     | 0.28    |     | -0.0023               | 0.57    |     |
| Blogger_Certification        | -0.3294     | 0.25    |     | 0.0013                | 0.75    |     |
| Blogger_GradSchool           | 0.7350      | 0.03    | **  | 0.0080                | 0.09    | *   |
| Blog_Institutional           | 0.3629      | 0.14    |     | 0.0037                | 0.29    |     |
| Tech_Score <sup>1</sup>      | -0.0060     | 0.17    |     | <.0001                | 0.90    |     |
| Investing                    | -0.2609     | 0.14    |     | 0.0027                | 0.27    |     |
| Mean_MarketCap <sup>1</sup>  | <.0001      | 0.37    |     | <.0001                | 0.57    |     |
| SCAR Pre-Event 20days        | 0.1620      | 0.02    | **  |                       |         |     |
| Abn Vol Pre-Event 20days     |             |         |     | 1.2558                | <.01    | *** |
| <b>R<sup>2</sup></b>         | 12.0%       |         |     | 31.2%                 |         |     |
| <b>N</b>                     | 193         |         |     | 193                   |         |     |

<sup>1</sup> Coefficient Estimates are multiplied by 10<sup>5</sup>

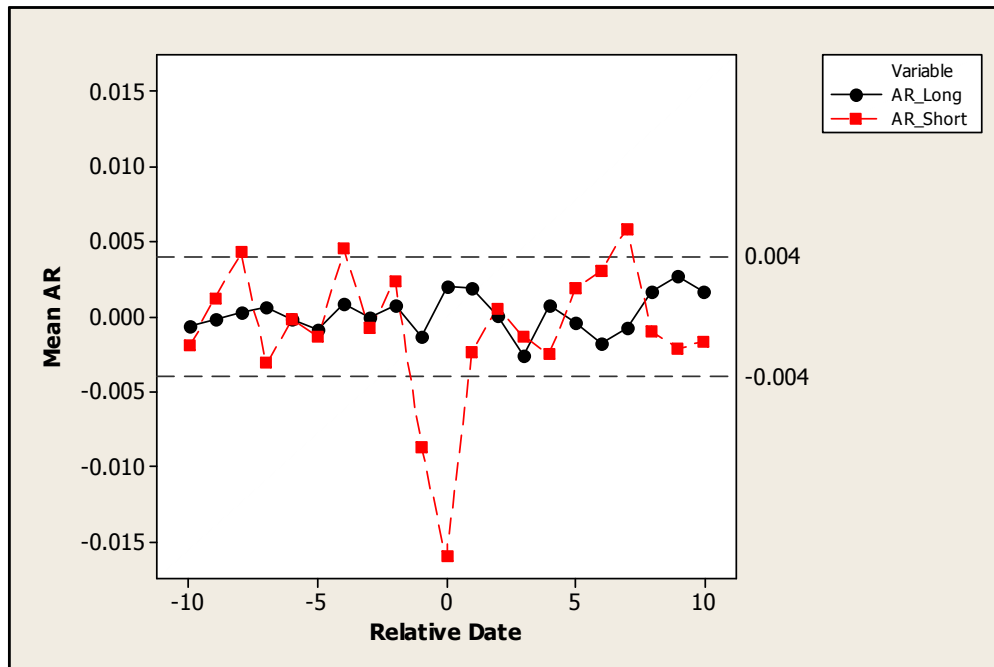
Cross-sectional regressions of standardized cumulative abnormal returns and abnormal trading volumes versus the hypothesized predictors (please note that SCARs for the short subset have been multiplied by -1). Dates are relative to the date of blog publication (the “Event Day” or date 0); hence, the “2-day Event Window” indicates cumulative results for day 0 and day +1. The table reports coefficient estimates, *p-values* relative to *t-tests* for significance of individual factors and overall *R-squared* statistics for the regressions:

$$SCAR\_2days_i = \alpha_i + \beta_1 Short_i + \beta_2 Multiple\_Recs_i + \beta_3 Number\_Of\_Words_i + \beta_4 Blogger\_Anonymous_j + \beta_5 Blogger\_Pro_j + \beta_6 Blogger\_Student_j + \beta_7 Blogger\_Journalist_j + \beta_8 Blogger\_Certification_j + \beta_9 Blogger\_GradSchool_j + \beta_{10} Blog\_Institutional_k + \beta_{11} Tech\_Score_k + \beta_{12} Investing_i + \beta_{13} Mean\_MCap_m + \beta_{14} SCAR\_PreEvent\_20Days_i + \varepsilon_i$$

$$AbnVol\_2days_i = \alpha_i + \beta_1 Short_i + \beta_2 Multiple\_Recs_i + \beta_3 Number\_Of\_Words_i + \beta_4 Blogger\_Anonymous_j + \beta_5 Blogger\_Pro_j + \beta_6 Blogger\_Student_j + \beta_7 Blogger\_Journalist_j + \beta_8 Blogger\_Certification_j + \beta_9 Blogger\_GradSchool_j + \beta_{10} Blog\_Institutional_k + \beta_{11} Tech\_Score_k + \beta_{12} Investing_i + \beta_{13} Mean\_MCap_m + \beta_{14} AbnVol\_PreEvent\_20Days_i + \varepsilon_i$$

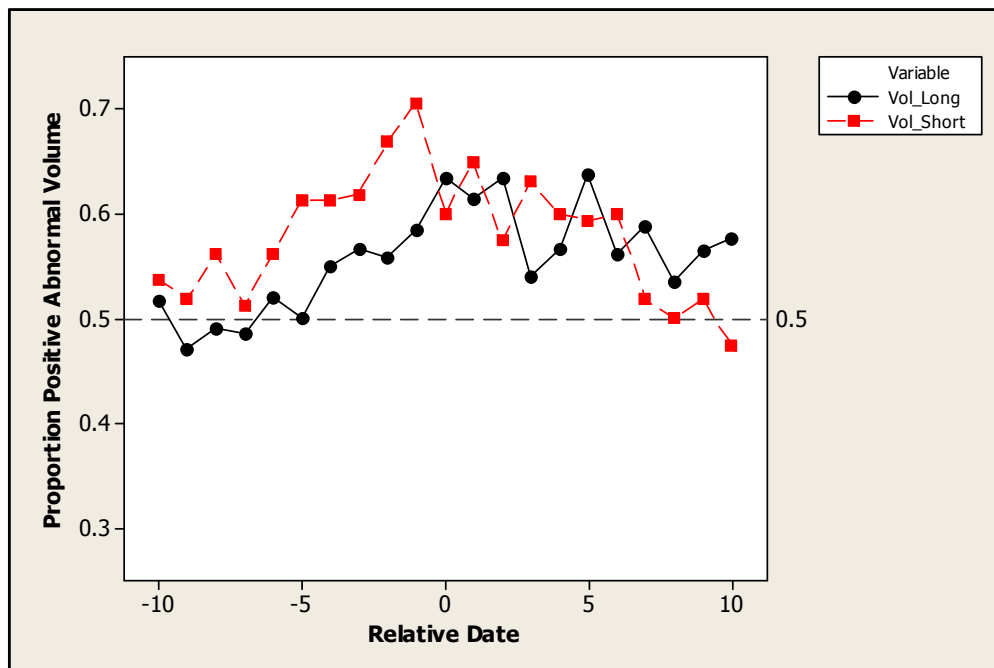
Significance is denoted as follows: “\*” indicates significance at the .1 level; “\*\*\*” indicates significance at the .05” level; “\*\*\*\*\*” indicates significance at the .01 level.

**Figure 1: Mean Abnormal Returns Vs Relative Date, All Data**



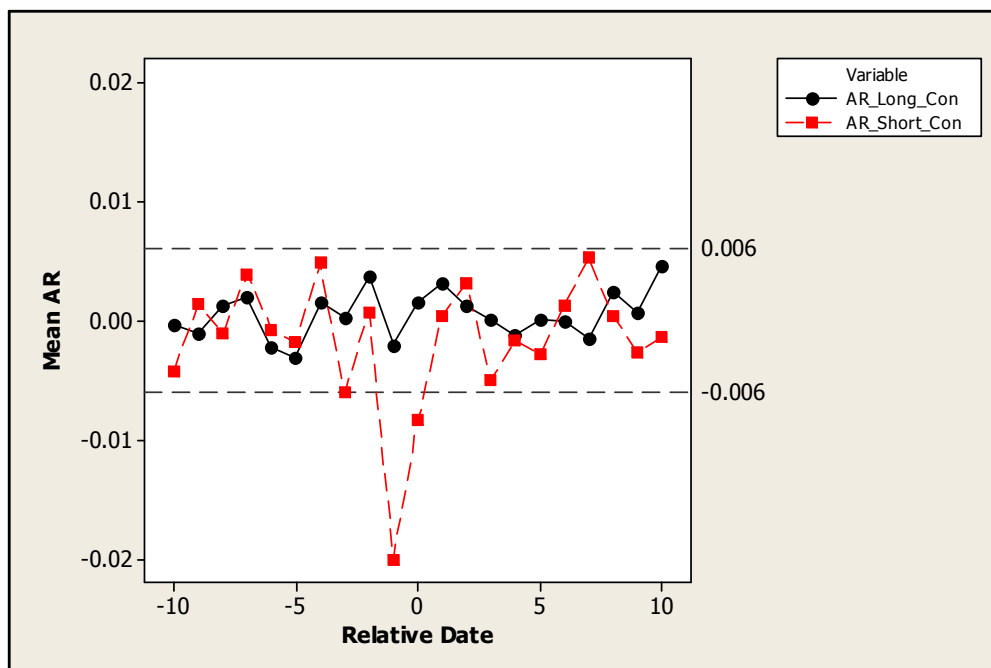
The above scatterplot presents mean abnormal returns for all days in the interval -10 to +110 for the buy (long) and sell (short) subsets. The horizontal lines are drawn at about two standard deviations from zero.

**Figure 2: Abnormal Trading Volumes Vs Relative Date, All Data**



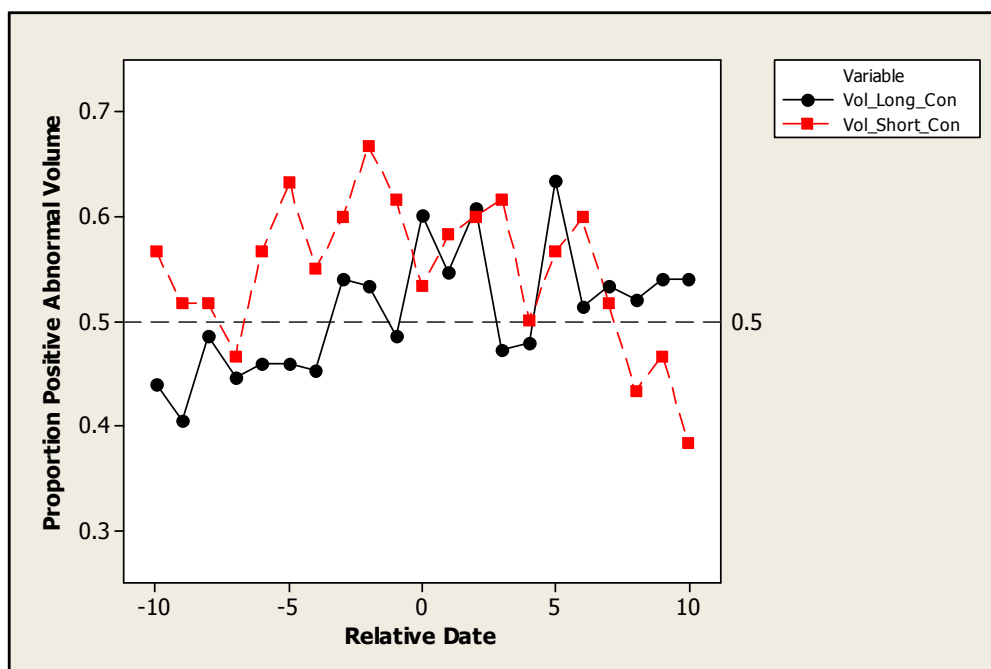
The above scatterplot presents the proportion of firms displaying positive abnormal volumes for all days in the interval -10 to +10, for the buy (long) and sell (short) subsets.

**Figure 3: Mean Abnormal Returns Vs Relative Date, Controlled Subset**



The above scatterplot presents mean abnormal returns for all days in the interval -10 to +10, for the buy (long) and sell (short) controlled subsets. The horizontal lines are drawn at about two standard deviations from zero.

**Figure 4: Abnormal Trading Volumes Vs Relative Date, Controlled Subset**



The above scatterplot presents the proportion of firms displaying positive abnormal volumes for all days in the interval -10 to +10, for the buy (long) and sell (short) controlled subsets.